ABSTRACT

The study entitled "**Floodline Alert: An Arduino-Based Flood Warning with SMS Notification**" was conducted to address the high vulnerability of low-lying barangays in Guipos, Zamboanga del Sur, to river overflows. Currently, the lack of timely flood warnings often results in delayed evacuations and significant loss of property and livelihoods. The primary objective of this project was to develop a reliable, cost-effective flood monitoring system for residents and the Local Disaster Risk Reduction and Management Office (LDRRMO). Specifically, the system aims to accurately detect rising water levels and automate the dispatch of SMS alerts to registered users without requiring internet access. The researchers adopted the 4D Model (Discover, Design, Develop, Deliver) as the development framework, integrating an ESP32 microcontroller with a dual-sensor unit consisting of analog sensors and float switches. The system was developed using web-based technologies for the admin dashboard and an HTTPSMS gateway for communication, supported by solar panels and power banks for continuous operation during power outages. Technical evaluations through eight test cases, including stress and safety testing, confirmed the system's stability and its ability to provide a failsafe 6-minute automated SMS dispatch. Results showed that the device effectively handles abnormal conditions and maintains an auditable record of all sensor readings. The implementation of this system provides an accessible disaster risk reduction tool that empowers the community to respond effectively to flood threats, ultimately enhancing public safety and resilience in the municipality.

Keywords: *flood warning system, ESP32, SMS notification, 4D Model, disaster preparedness, Guipos Zamboanga del Sur*